Figure 7: Sync interface details

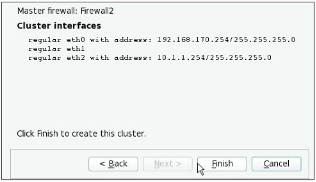

Figure 8: Finish cluster object

Since the entire *fwbuilder* package is GUI-based, make sure you start the GUI mode on your firewall manager machine. Now open a terminal window, and type *fwbuilder*, which should launch the *fwbuilder* console.

Firewall Builder v 4.0 natively supports different state sync groups, including *conntrack*. That means you don't need to explicitly add any rules to cover VRRP and sync traffic. If you are using a version other than this, please refer to the *fwbuilder* documentation (http://www.fwbuilder.org/4.0/docs/users_guide/) to enable access for VRRP and the *conntrackd* daemon.

Creating a firewall under *fwbuilder* is pretty straightforward. Let's now set up the cluster in Firewall Builder. Open *fwbuilder*, right-click on a firewall, and select *New Firewall*. The screenshot in Figure 3 depicts this; I have created both my firewalls under the *Firewalls* section.

After this, we need to create a firewall cluster, using the firewalls we just created. To do so, right-click on *Clusters* and choose *New Cluster*. See Figure 4.

Provide a cluster name, and select the firewalls that will be a part of this cluster. Also choose which one will play the master role (see Figure 5).

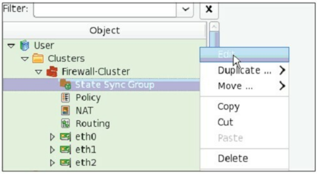

Figure 9: State sync group

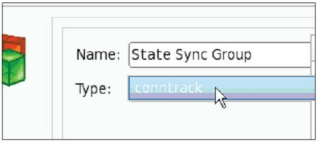

Figure 10: Choosing state sync protocol

Next, select the *Cluster Interface* name, enter the *Label* name, click on *Add Address*, and provide the VRRP IP address. Select VRRP as the protocol from the drop-down menu. Repeat this step for the entire cluster's interfaces, and add VRRP IP addresses for all those interfaces.

For *Sync Interface*, you must select *None* as a protocol. See Figure 7.

Review the configuration once again, and click *Finish* to create the cluster, as in Figure 8.

Once you are finished with this, expand the *Cluster* object, and edit '*State Sync Group*', as in Figure 9, and select *conntrack* from the drop-down, as shown in Figure 10.

That's it! You are done with the cluster configuration; now it's time to add the rules, and push the policy. Before that, ensure that you create a */etc/fw* directory on every cluster member, where the deployed policy will be stored. Figure 11 is a snapshot of my sample rule set.

Let's now deploy the policy. For that, you need SSH and SCP. Locate the appropriate binaries, as shown in Figure 11, and click *OK*. (If your console is deployed on Windows, you need *plink.exe*.)

Click *OK*, and click the *Compile Policy* button. It automatically detects rule shadowing, informs you if any error is found, and if not, pushes the policy to both the firewalls.

In the end, log in to each cluster member, and save the *iptables* policy with the *service iptables save* command.

Final testing

So far, our cluster is up, and previous tests have confirmed VRRP fail-over works. Let's now carry out final end-to-end